

CHAPTER

2

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2.1 Installation Precautions

In order to fully exhibit the functions of the checkweigher, maintain precision, and perform weighing safely, the installation site should conform to the following conditions:

■ Ambient Temperature and Humidity

- Do not install the checkweigher to a place which may become extremely hot or cold, or exposed to direct sunlight.
- Temperature range: 0°C to 40 °C
- Humidity range: 30% to 85% (There should be no condensation.)

■ Installation Surface

Install the checkweigher on hard and horizontal surface with no vibration.

■ Installation Space

Install the checkweigher so that there is sufficient access around it to perform maintenance, inspection, etc.

■ Electrical Power

- As specified at order time.
- Carry out wiring so that the voltage does not vary more than $\pm 10\%$ under load.
- Do not connect power devices, which may emit electrical interference vibration, to the same power line as the checkweigher.
- Be sure to connect a ground to the outlet and carry out the third class grounding. Insufficient grounding may cause an electric shock due to electric potential difference when touching a machine and another.

■ Static Electricity

Be sure all parts are fitted securely and that all metal parts are grounded.

■ Electromagnetic Interference

Do not install the checkweigher near any devices which may cause electromagnetic interference, etc, eg., a wireless transmitter.

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2.2 Installation Procedures

2.2.1 Unpacking work

Remove tape, vinyl sheet, and cushioning materials from the main body, and take out the parts packed in the small box.

(In some cases, the checkweigher is packed in a crate. In this case, after removing the crate, take out the main body, rejector device, and the small box containing parts, and proceed with the unpacking work.)



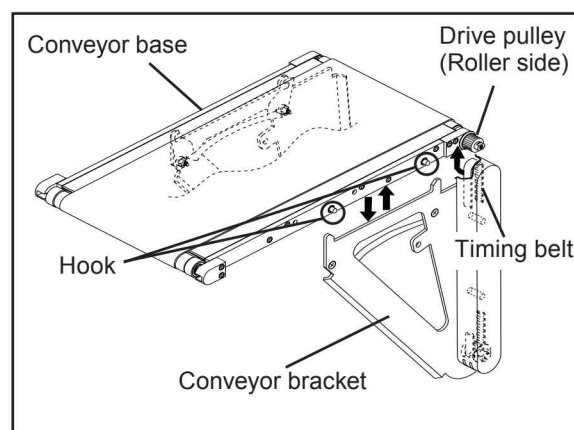
Do not place a heavy object on the weigh conveyor or apply excessive stress during operation. Doing so may damage the weigh mechanism (load cells).

2.2.2 Assembling work

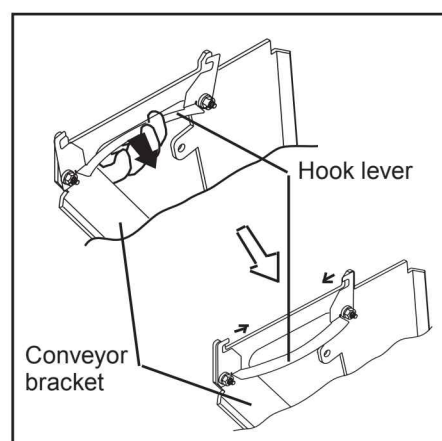
Place the DACS-GN main body and the RE-015/060 rejector device (option) on the installation site, and carry out the assembling work.

■ Installation of weigh conveyor (SS type)

1. Route the timing belt over the pulley.



2. Place the conveyor base hook on the securing section of the conveyor bracket, and lightly pull down the plate spring with the fingers as shown on the right figure. Then the lever hooks move in the central direction. The conveyor base can be now secured.



NOTE

Secure the conveyor base to the conveyor bracket after hanging the driving pulley (roller side) on the timing belt.



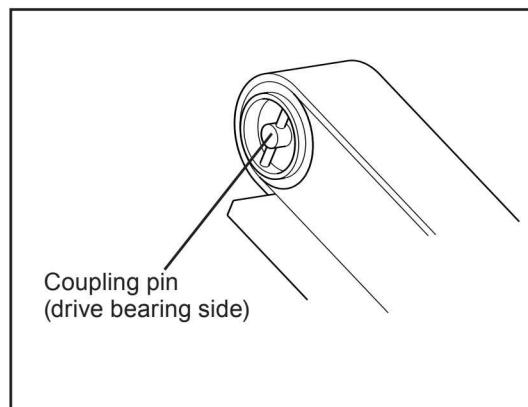
Do not install the weigh conveyor by applying excessive stress. Doing so may damage the weigh mechanism (load cells).

■ Installation of weigh conveyor (CR type)

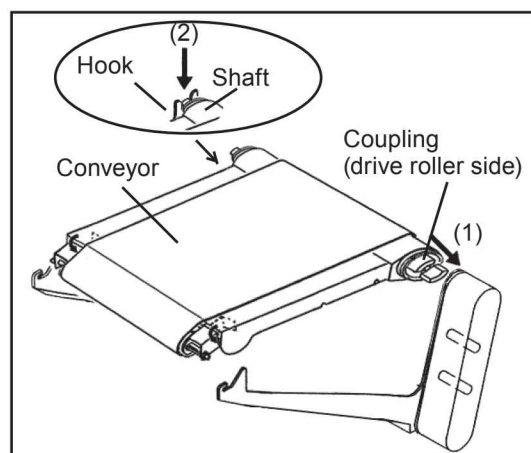
- Align the coupling notch of the drive roller side of the weigh conveyor with the coupling pin of the drive bearing side.

NOTE

In assembling the drive roller coupling, ensure correct setting; the coupling pin may be hard to fit the notch of weigh conveyor.



- Insert the conveyor (drive roller) at the discharged side of the weigh conveyor to the direction (1), then to the direction (2).

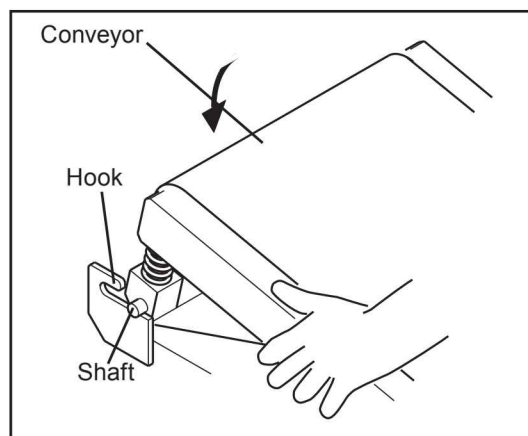


Be sure to hold the conveyor bracket by your fingers and then push the conveyor bracket so as not to get your finger caught.

- Hold the intake side of the weigh conveyor by gripping the conveyor bracket as shown in the right figure and press the infeed conveyor. (Easy disassembly structure helps you to attach it easily.)

NOTE

Do not apply excessive load to the weigh conveyor. The weigh conveyor is directly connected to the weigh sensor.



■ Installation of the infeed conveyor (SS/CR)

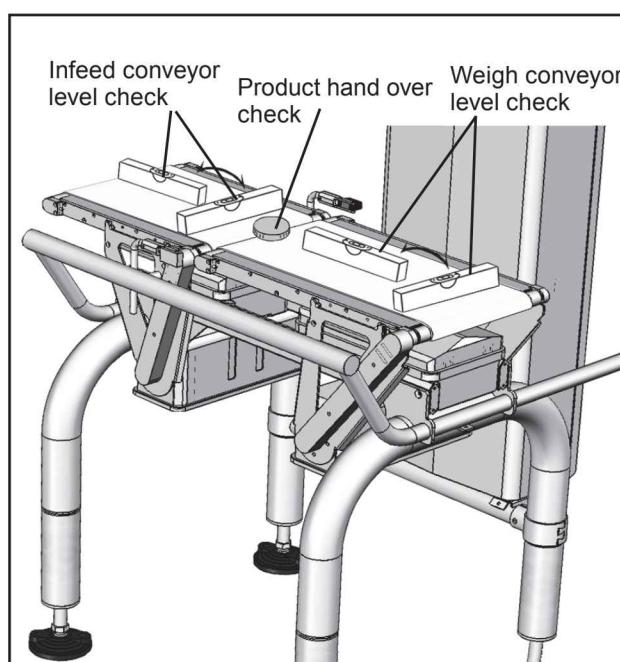
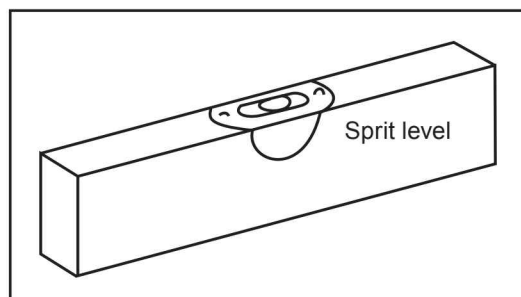
Refer to the installing procedures of each weigh conveyor for installation.

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2.2.3 Installation work

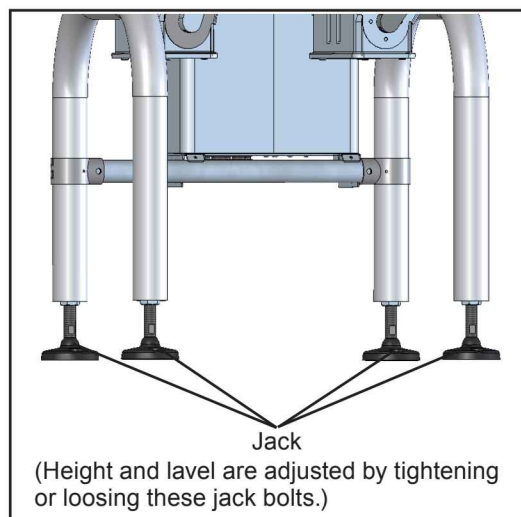
Put the main unit and rejector (option) with a suitable position.

1. Using a spirit level, adjust the height and level by tightening or loosening the jack bolts until the weigh conveyor transfer surface of the DACS-GN main body becomes horizontal.
2. Adjust the jack bolts so that there is no gap or difference in height between the infeed conveyor and the previous process conveyor, and make sure that the jack bolts keep absolute Spirit Level contact with the floor.
3. By using a level, adjust the jack at the infeed conveyor so that the flow direction and the width direction on the infeed conveyor are parallel.
4. Confirm that the product is passed smoothly from the infeed conveyor to the weigh conveyor.
5. Tighten the lock nuts of the jacks.



NOTE

- If there is a wide gap or difference in height between the conveyors, the impact force will adversely affect the precision when products are transferred to the infeed conveyor. When the jack bolts do not keep absolute contact with the floor, the main body will rattle, which may adversely affect the precision.
- Never fail to check that the infeed conveyor as well as the weigh conveyor is set horizontally while adjusting them by 4 jack bolts. Adjusting 1 leg forcibly will cause a difference between the surface levels of two conveyors.



6. Set the rejector (optional), and move the jack to adjust height and levelness so that the transfer surface of rejecting conveyor becomes horizontal.
7. Adjust the jack so that there is no gap or difference between the rejecting conveyor and the weigh conveyor. Also confirm that there is no jack floating.
8. Tighten the lock nuts of the jacks.



Conveyors must keep a certain distance between them. Large clearance or any level differences may cause poor product hand-over, leading to a rejecting error due to timing gap.



REFERENCE

For detailed operation of the rejector, refer to Chapter "9 REJECTOR" .

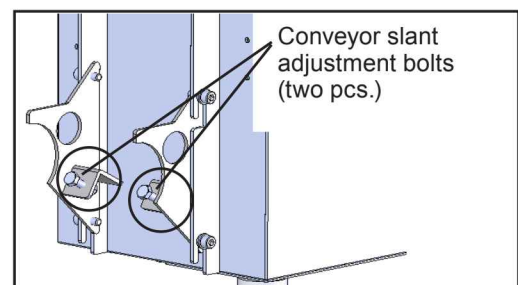
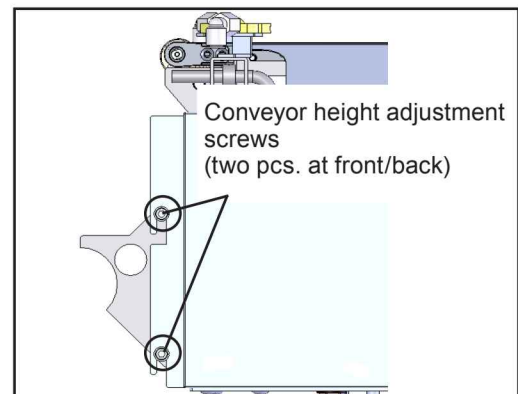
2.2.3.1 Connecting rejector with DACS-GN main unit

■ Air-jet type

Air jet type rejector is delivered connected with the DACS-GN main unit.

To fine adjust the conveyor height, loosen two screws fixing the rejector. To adjust the conveyor inclination, adjust the adjusting bolts of connecting fitting.

1. Loosen the lock nut.
2. Turn the inclination adjusting bolt to adjust the inclination of the conveyor.
3. Tighten the lock nut after adjustment.



When elevating the lower part of the conveyor, turn the bolt while lifting the lower part of the rejector slightly up so as to decrease the pressure on the bolt. It may damage the bolt if turning it forcibly.

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■ Arm type

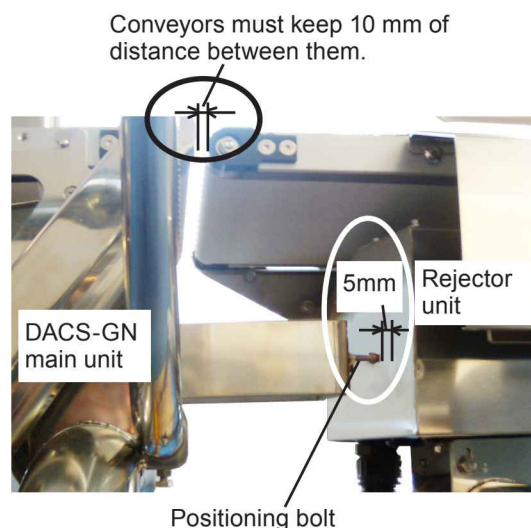
Two rejector-positioning bolts are attached to the special bracket of the DACS-G main unit. Place the rejector approx. 5mm to the positioning bolts of the main unit from its drive side surface.

At this time, make sure the rejector and weigh conveyor of the main unit keep 10mm distance. Distance can be adjusted by the positioning bolts.



Be sure that the stopper has no contact with the rejector. When the stopper contacts the rejector, vibration caused by the rejector could result in poor weighing accuracy.

1. Loosen the lock nut.
2. Turn the positioning bolt in a counterclockwise direction to move the rejector away.
3. Tighten the lock nut.



■ Drop belt type

Set the rejector at the lower part of the DACS-GN main unit and align the height of the rejector with that of the conveyor.

Make sure that the rejection conveyor does not touch the weigh conveyor.

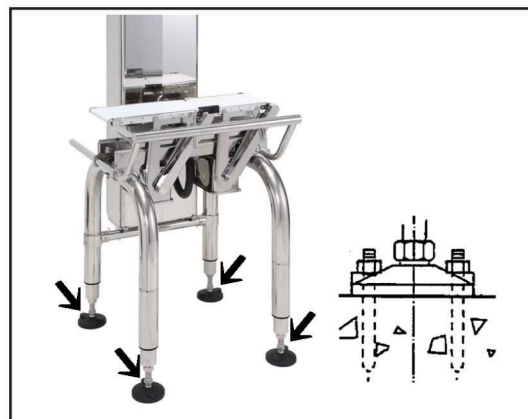


2.2.4 Fixation of rejector



When the specifications of the checkweigher require fixation by anchoring or if there is a risk that the equipment will topple over, secure the DACS-GN main body and the 015/060 rejector device to the floor with screws, etc.

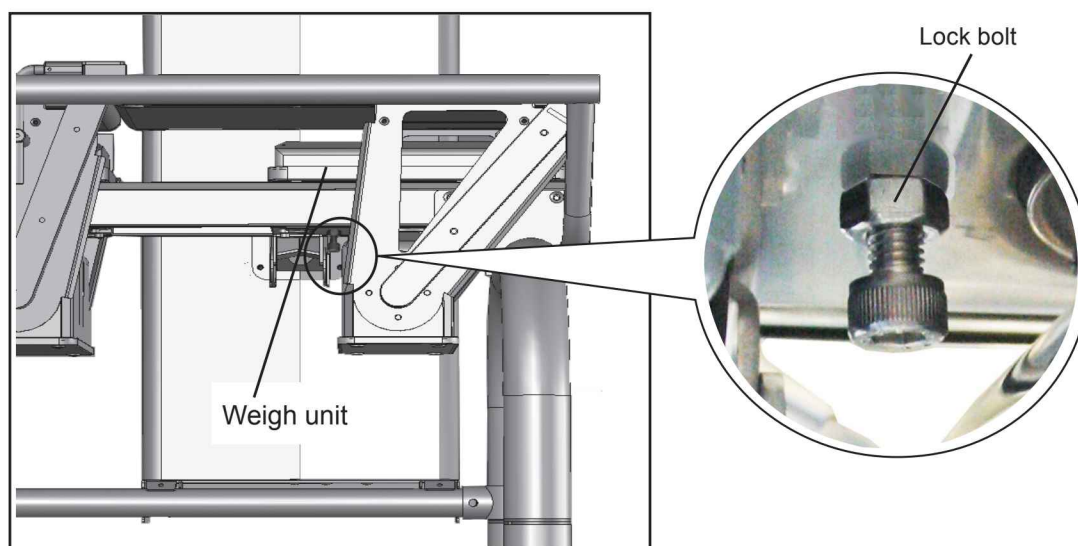
1. As shown in the figure, bore holes with $\phi 8.5$ and 35 mm in depth in the floor.
2. Pound a concrete anchor suitable for the hole size into the hole.
3. Secure the legs of the main body to the floor by tightening the concrete anchor with M10 nut.



2.2.5 Removal of weigh unit protective lock bolt

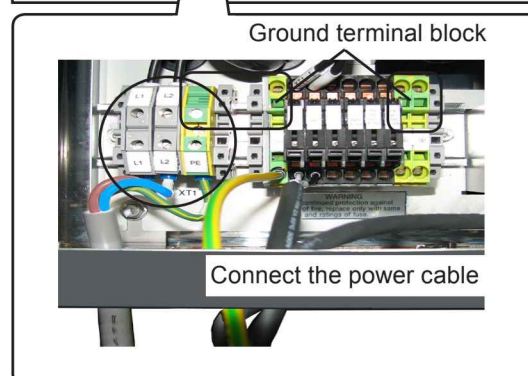
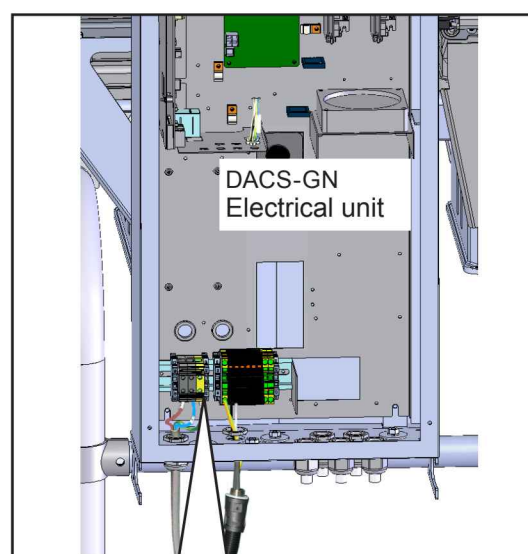


The checkweigher fixes the weighing mechanism using the lock bolts to protect the mechanism during transportation. After completing the installation, ensure to remove the lock bolt and mount the plug bolt to start the operation.
(For the model of twin-cell specification, there is a lock bolt at the infeed side.)



2.2.6 Wiring work

1. Remove the cover at the back of electrical unit, and draw the power cable inside the electrical unit from the opening at the bottom of the unit.
2. Connect the power cable to the terminal block: XT1-L1, 2, PE (chassis ground).



- The PE (ground) terminal board at the main unit grounds the chassis when installed, and no special wiring from the ground terminal to the chassis is provided.
- For the specification outside Japan, the color of power cable is "Brown, blue, green/yellow." Connect the brown cable to the terminal block XT1-L1, the blue cable to L2, and the green/yellow cable to PE (chassis ground).

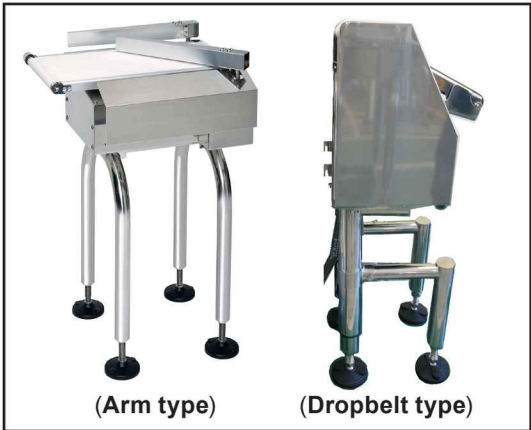
2 SETUP

2.2.6.1 Rejector [RE-0xx-A3 (Arm), DB2 (Dropbelt)]

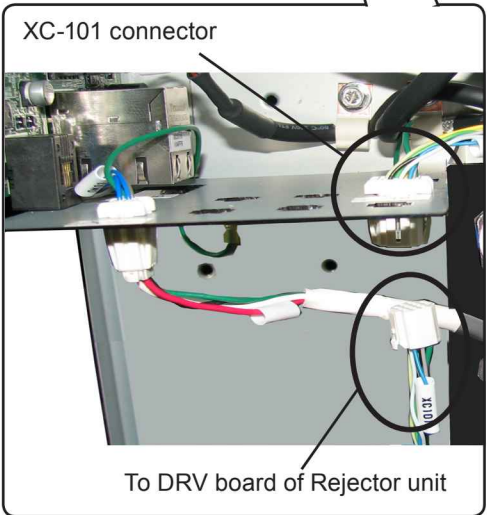
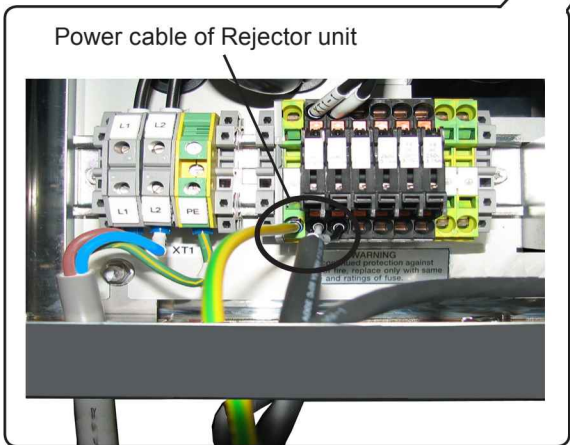
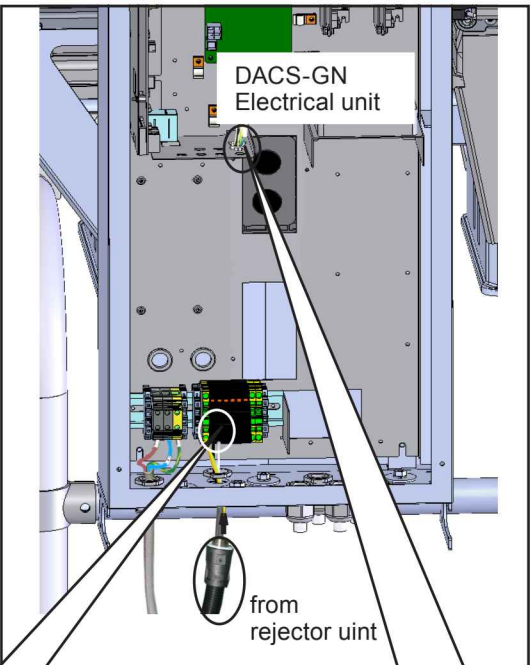
1. The main part of the reject unit is fixed to the main unit of DACS-GN. It is shipped from the factory with the connection cable (flexible tube) connected.



The RE-0XX side has been already connected in the factory default status.



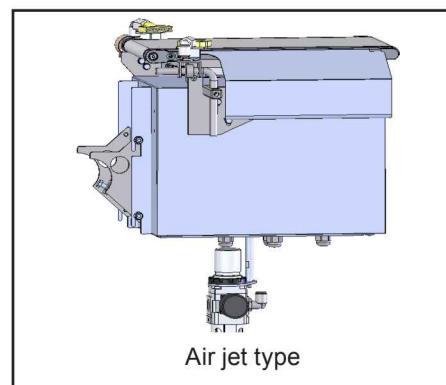
2. Remove the cover at the rear of instrumental unit, pull the connection cable (flexible tube) from the opening at the bottom of instrumental unit to inside the instrumental box, and fix the flexible tube at the bottom.
3. Connect the cable connector for reject control signal (XC101), and connect the power cable to the terminal block XT1-1,2 and connect the grounding cable to the terminal block XT1-grounding.



2.2.6.2 Rejector [RE-0xx-J3 (Air jet type)]

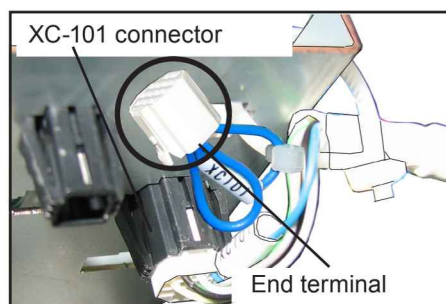
As in the right photograph, the main unit of RE-0XX-J3 rejector is fixed to the main body of DACS-GN.

The connection cables (flexible tubes) has been connected in the factory default status.



NOTE

When there is no reject unit, or when the main unit should be operated with the reject unit unconnected for a test purpose, be sure to connect the end terminal connector to the XC-101 relay connector. Otherwise, a connection error will occur in the reject unit, disabling the operation.



2.2.6.3 Connecting DACS-GN machine to external devices (input/output signal specification) (optional)

The input/output signal is option. Adding a set of option means adding four contacts for each of input and output signals, to which contact the type of input/output signal can be selected and set via the remote control. Either "NO" contact or "NC" contact can be selected according to the connection method.



REFERENCE

Refer to "4.7 Relay Terminal Unit (optional)" for the detail of Input/Output Function.



TIP

In the UL specification, the terminal connector is stored in the tool box.

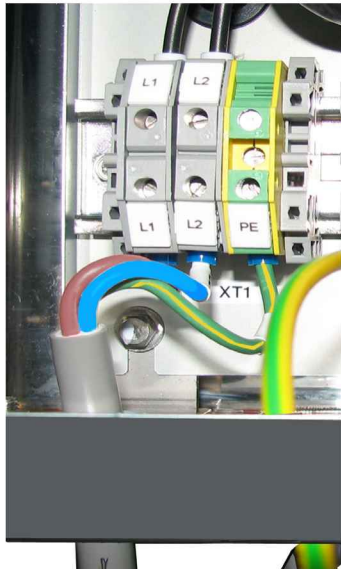
2 SETUP

2.3 Post-Installation Check

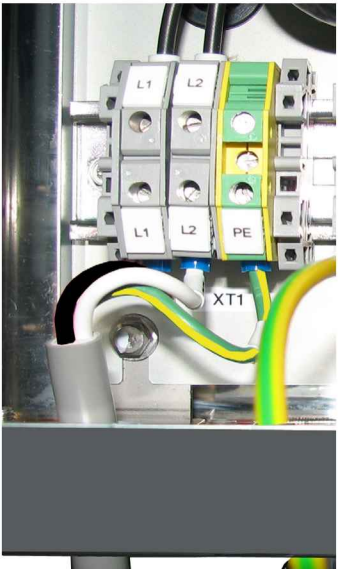
2.3.1 Connecting main power supply [UL type]

NOTE The power outlet should be placed around the machine (DACS-GN main body).

■ Main power supply terminal (XT1)



[Input power supply: 100V to 125V]



[Input power supply: 200V to 250V]

2.3.2 Connecting main power supply [EU type]



[Input power supply: 200V to 240V]

■ UL type

- XT1 (L1/L2/PE)

Conductor cross section flexible max.	2.5 mm ²
Conductor cross section AWG max.	14
Connection in acc. with standard	DIN 46234 DIN 46235 DIN 46237
Tightening torque, min	0.6 Nm
Tightening torque max	0.8 Nm

- XT1 (1-6)

XT1 (1-6)	
Conductor cross section solid max.	6 mm ²
Conductor cross section flexible max.	4 mm ²
Stripping length	10 mm ... 12 mm

- XT101-104 (31-42)

Interface with external equipment, connect power supply from SELV
(safety extra low voltage)
circuit within following maximum rating

Maximum Rating of Output Signal

Item \ Load	Resistance Load ($\cos\phi=1$)	Inductive Load ($\cos\phi=0.4$ L/R=7ms)
Rated Load	DC 30V 5A	DC 30V2A
Rated Current	5A	5A
Max Contact Current	5A	2A

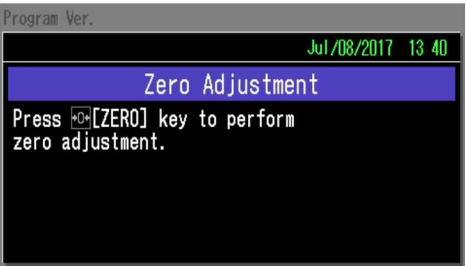
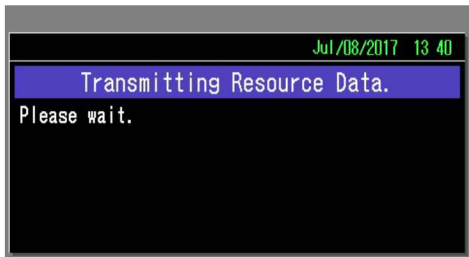
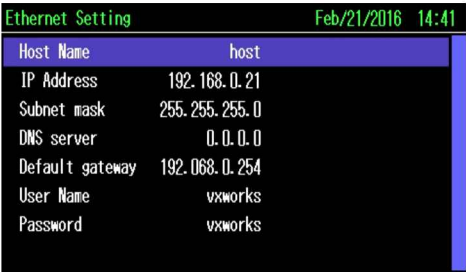
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2.3.3 Turn the Power On

Turn the power switch "ON" to show the ISHIDA logo mark on the display. (It takes 40 seconds). Then, a program number display, a preset data checking display, and a zero adjustment standby display will appear in order.



At the first restart after the software is installed, a screen on the "Ethernet setting" appears. Usually press  key without changing any parameter.



NOTE

- Setting [Auto Zero Adjustment] in the [System Configuration] to "Yes" enables [Auto Zero Setting] operation without displaying the [Zero Adjustment] message and then displays the operation standby screen. (For standard setting, the screen indicated above is displayed)
- If a communication error occurs when the program number display appeared, the problem may be poor connection or a defective MCU, DRV, ADC board. Check the connection area and the LED on the board.

■ Operation key arrangement

The operation keys to use for setting/adjustment are encircled by white lines in this manual.



REFERENCE

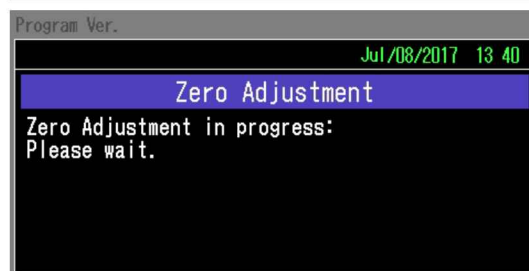
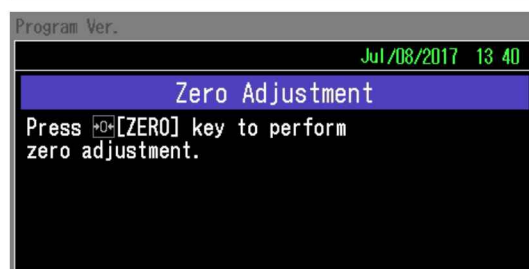
For detailed function of each operation key, as well as operation and function of command dial, refer to "OPERATION PANEL" and "OPERATION" in the Instruction Manual.

2.3.4 Zero Adjustment

1. After confirming that there are no products on the weigh conveyor, press the $\rightarrow 0 \leftarrow$ key on the zero adjustment standby menu.
 - Zero adjustment starts.
 - The "Zero Adjustment in progress" message appears.
 - When zero adjustment is completed, the production standby menu appears.
2. Confirm that weight display indicates 0.0 ± 0.2 g. (at weigh capacity: 1.5 kg)
3. If the weight display does not indicate 0.0 ± 0.2 g, press the $\rightarrow 0 \leftarrow$ key again to perform zero adjustment.



Operation keys
Operation/Display panel



NOTE

- If the weighing capacity is set to 600 g, the display shows 0.00 g.
- If the weighing capacity is set to 1.5 Kg, the display shows 0.0 g.
- The minimum quantity to scale when the weighing capacity is 600 g is 0.05 g.
- The minimum quantity to scale when the weighing capacity is 1500 g is 0.1 g.
- If the weighing capacities are 3000 g and 6000 g, the display shows 0.0 g.
- The minimum quantity to scale when the weighing capacity is 3000 kg is 0.2 g.
- The minimum quantity to scale when the weighing capacity is 6000 g is 0.5 g.
- When the checkweigher is stopped, zero adjustment is available by pressing the $\rightarrow 0 \leftarrow$ key during the operation standby screen is displayed.
- When the tolerance value for automatic zero setting is set to 0 g and turning the automatic zero setting to OFF, the zero point display takes 15 to 20 minutes to stabilize after the power ON. Set the tolerance value for automatic zero setting to approximately 4.0 g.

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2.3.5 Changing of Operational Level



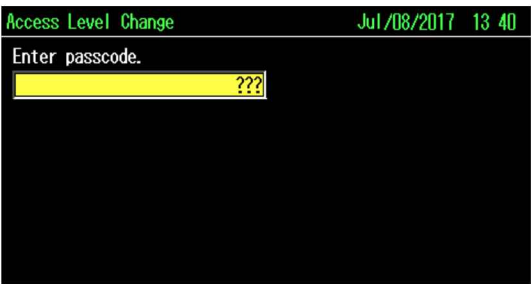
1. Press the  key on the production standby menu.
 - The Setup Menu of the operational level appears.
2. Point the cursor to "Access Level Change", and press  key.
 - The Access Level Change menu (Enter Password) appears.
3. Enter the password of the Level 3, and press the  key.
 - The Level 3 Setup Menu appears.

NOTE

- Contact to ISHIDA agent for the password required in the Maintenance Service level.
- When entering the password, the numbers are displayed as "???" on keypad.



Refer to the Operation Manual for the setting and changing procedure of passwords.

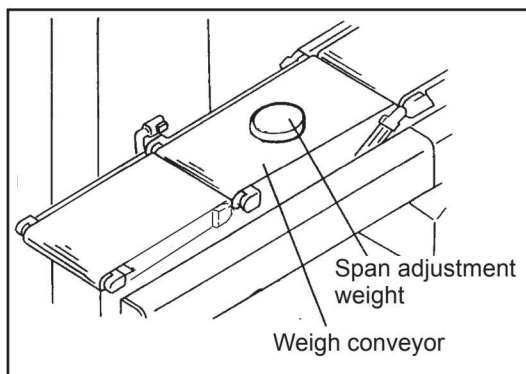
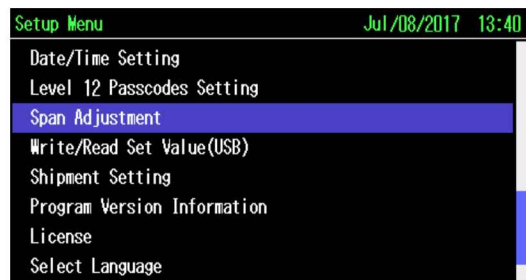


2.3.6 Span Adjustment (Standard type)

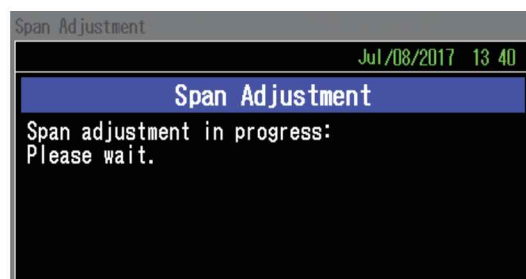
1. Point the cursor to "Span Adjustment", and press  key.
 - A selection screen for gram/ounce display appears.
2. Select the gram display option and press the  key.
 - If the destination is not US, this selection screen does not appear on the screen.
 - The Span Adjustment menu appears.
 - The weight of the span adjustment weight (span weight) appears in the Span Adjustment menu.

NOTE The weight of the span adjustment weight differs depending on the model.

3. Place the attached span adjustment weight in the center of the weigh conveyor



4. Press the  key.
 - Span Adjustment starts.
 - The "Span adjustment in progress" message appears.
 - When span adjustment is completed, the "Span adjustment completed" message appears.
5. Confirm that the displayed weight indicates the same weight as the span adjustment weight



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6. Remove the span adjustment weight from the weigh conveyor, and confirm that the displayed weight indicates $0.0\text{ g} \pm 0.2\text{ g}$.
7. Press the  key twice.
 - The production standby menu returns.



NOTE When the span adjustment weight is 1 kg, the displayed margin of a span error is $\pm 0.1\text{ g}$ at the 1 kg weighing capacity, and $\pm 1.0\text{ g}$ at the 5 kg weighing capacity.

2.4 Preparation of Test Operation

2.4.1 Check the Belt Condition

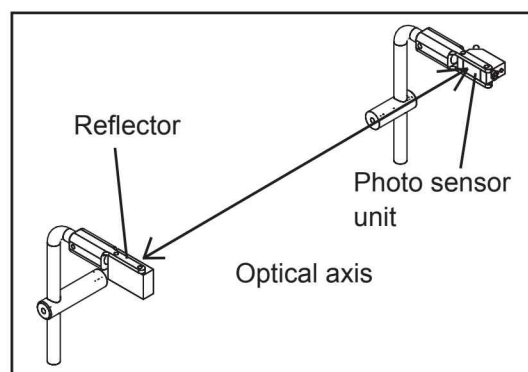
1. Confirm the belt tension of weigh conveyor.
 - Confirm that the belt of the weigh conveyor is free from excessive tension or slack.
2. Confirm that the belt of the weigh conveyor is free from lateral shifting (offset).
 - Press the  key to start the conveyor operation.
 - Carry on the conveyor operation for 15 to 30 min., and confirm that the belt of the weigh conveyor is free from lateral shifting (offset).
 - When the belts are free from lateral shifting (offset), press the  key to stop the conveyor operation.



Refer to "3.6.2 Tension and meandering adjustments of the weigh conveyor belt".

2.4.2 Check the Optical Axis of the Photo Switch

The optical axis of the photo sensor and the reflection plate needs adjustment. Adjust both photo sensor side and the reflection plate side so that only the green lamp lights with no axis interruption, and that the green lamp and the orange lamp light with the axis interruption.



2.4.3 Preparation for Test Run

■ System Configuration Setting

Set each item of "System Configuration" according to the model specifications.

■ Reject Setting

Set each item of "Reject Setting" according to the specifications.

■ Statistics & Print Setting

Set each item of "Statistics & Print Setting" according to the specifications.

■ Preset

Set each item of "Preset" according to the products of users.



Refer to the Operation Manual for the setting operation procedures of the "System Configuration", "Reject Setting", "Statistics & Print Setting", "Preset", etc.

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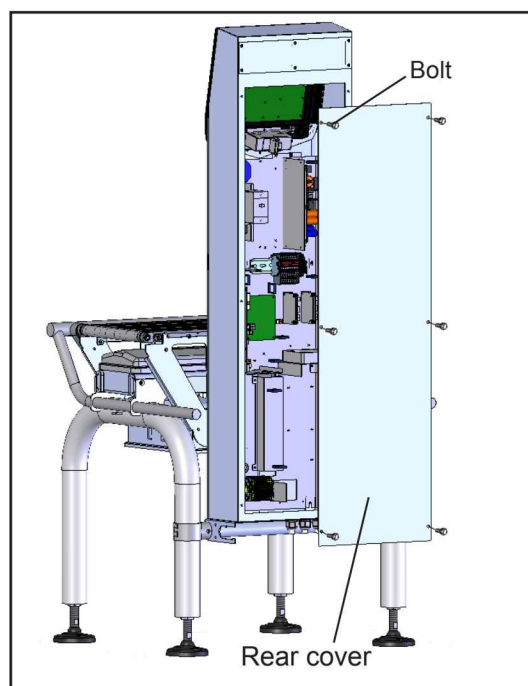
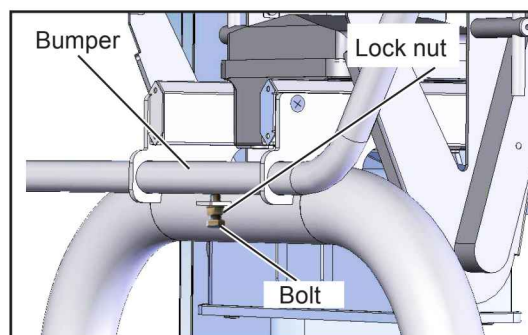
2.5 Procedures of Changing Operation Unit Position

This section explains how to change the location of the operation unit so that products flow from the right side of the operation screen.



■ Procedures

1. Turn OFF the power supply of the machine to cut off the air source.
2. Loosen both bolts fixing the bumper.
Loosen the nuts fixing the bolts first.
3. Remove the bumper.
4. Remove the 6 bolts on the rear side of the machine and detach the rear cover.



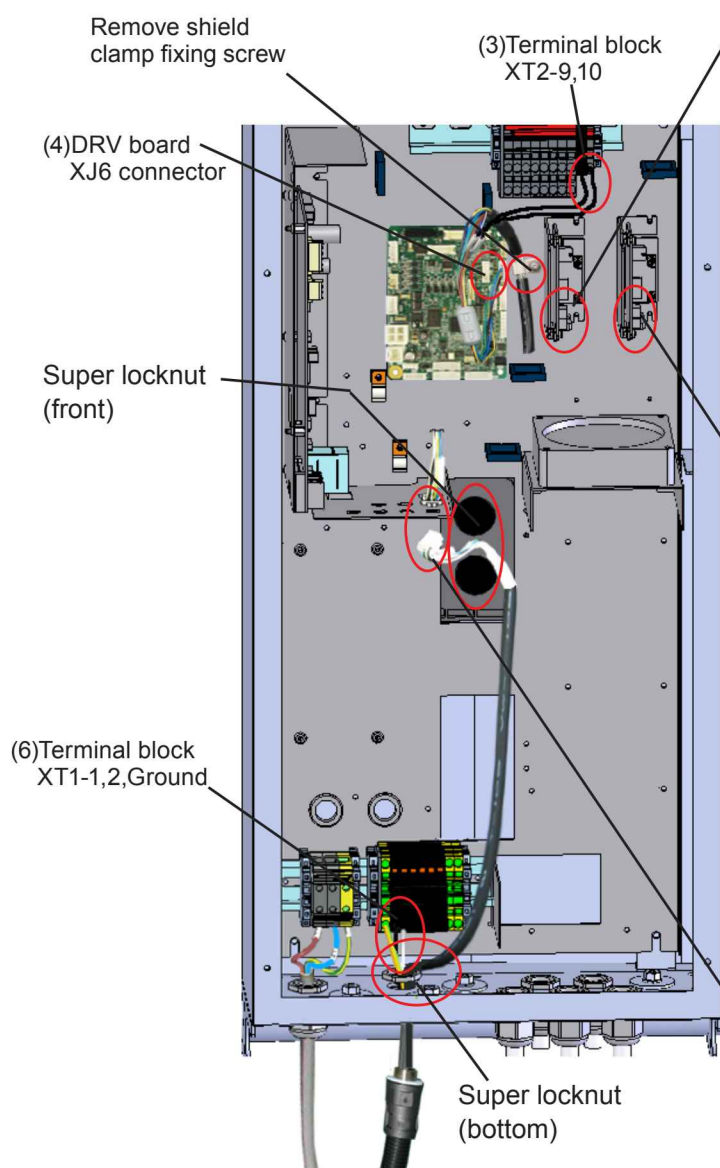
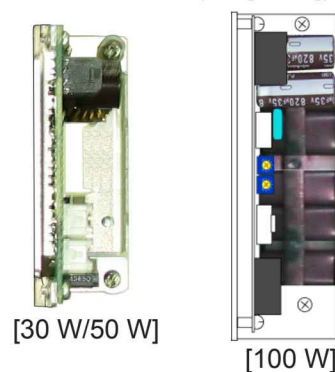
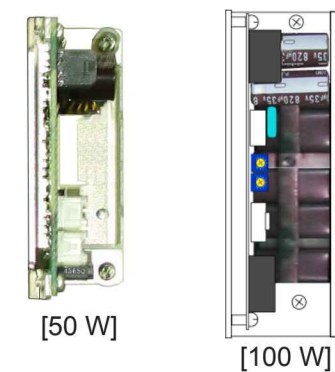
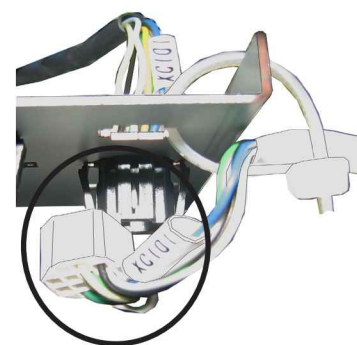
5. Disconnect wires below

Wires from the Main Unit

- 1) Controller connector: from the U3 IFC board: CN3.CN4
- 2) Driver connector: from the U4 WTC board: CN3.CN4
- 3) Power wire from ADC board (XJ5): XT2-9, 10 on terminal block
- 4) Signal wire from ADC board (XJ6): XJ5 connector on DRV board

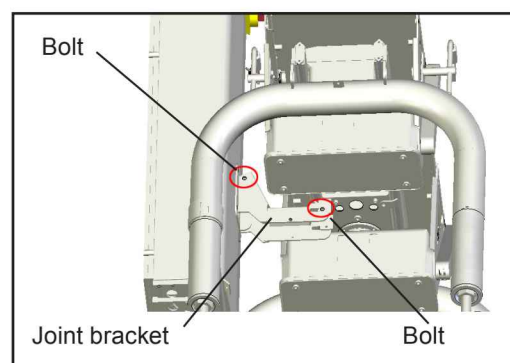
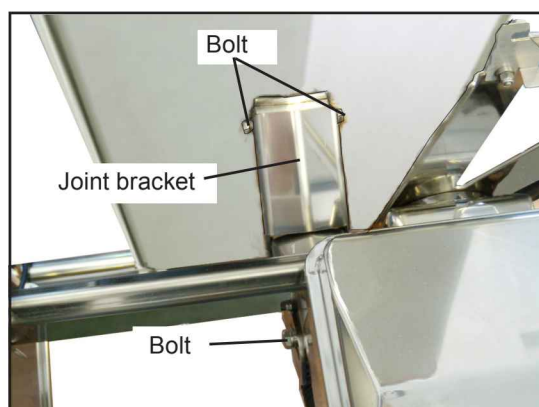
Wires from the Rejector

- 5) Signal wires: junction connector (XC-101)
 - 6) Power wires: terminal block XT1-1,2/XT1-Ground
6. Remove the super lock nuts of the flexible conduit connection inside the electrical control unit.
- Wires from the main unit: 2 places on the rear side
 - Wires from the rejector: at the bottom

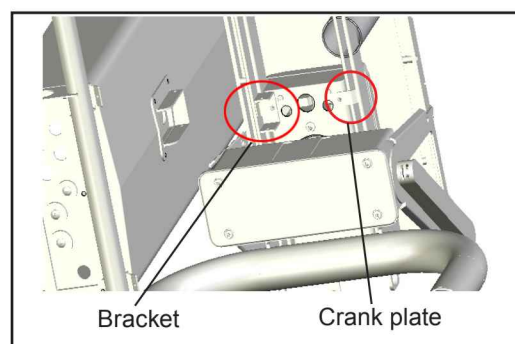
**Motor drive board (U4 [WTC])****Motor drive board (U3 [IFC])****(5) XC-101 junction connector position**

2 SETUP

7. Remove 4 bolts to disassemble the joint bracket connecting the electrical control unit and conveyor.

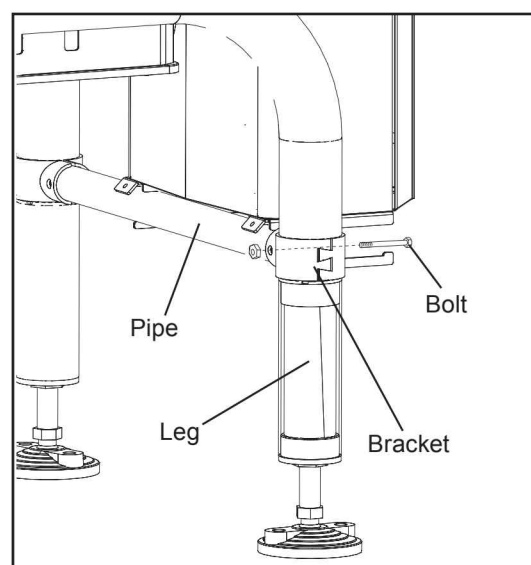


8. Remove 1 bolt to detach the bracket on the main unit to which the joint bracket was attached.



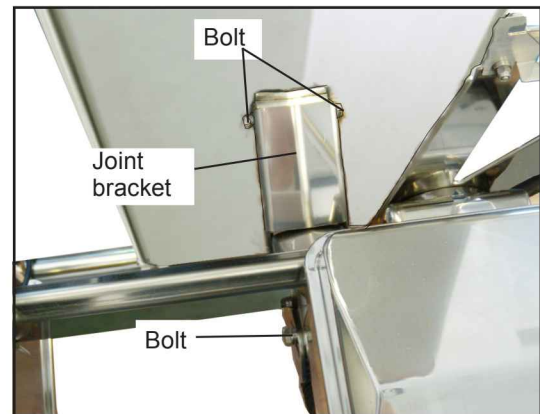
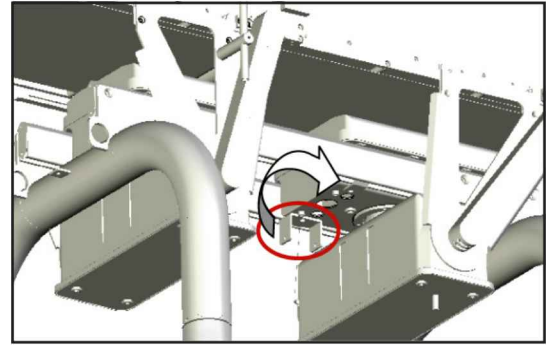
NOTE From this procedure of removing the electrical control unit, conduct with several people in order not to overturn or drop the electrical control unit.

9. Remove the right and left bolts fixing the pipe on which the electrical control unit is placed, and detach brackets fixing the legs.
10. Move the electrical control unit to the other side.
11. Attach the detached brackets on the other side of the legs. Make sure to adjust the position as high as the previous one. Connect the bracket and pipe on which the electrical control unit is placed.



NOTE If changing the mounting height of the control unit, it is necessary to change the joint bracket angle.

12. Insert a bolt to the attaching hole, adjust the mounting height of the electrical control unit, and tighten the nut temporarily.
13. Attach the bracket, which was on the main unit to which the joint bracket was attached, to the opposite side.
14. Tighten the super lock nuts of the flexible conduit connection.
15. Attach the joint bracket. Attach the joint bracket using a bolt to the conveyor side first, and attach the other side of the joint bracket to the electrical control unit while adjusting the side position of the electrical control unit.
16. Tighten the nut fully, which was tightened temporarily at the procedure 13.



NOTE Never fail to check that the electrical control unit is attached firmly. Otherwise, it may cause a drop or a fall.

17. Attach the bumper. Attach the bumper in an opposite direction and fix it by bolts. Additionally, tighten the lock nut to fix the bolts.
18. Reconnect the wires.
19. Restore the power and check the operation capabilities.
20. If it functions properly, attach the rear cover.

NOTE Be careful not to receive an electric shock while checking the operation capabilities.

2 SETUP

2.6 Procedures of Changing Conveyor Height

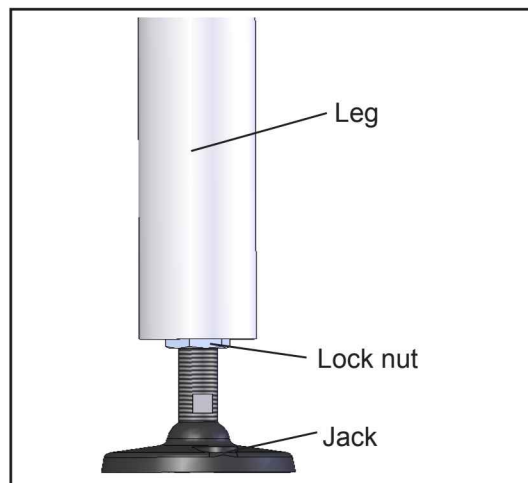
2.6.1 Standard model

Be sure to change the conveyor height within the height adjustment range of the jack.



To change the conveyor height in 100 mm or more, the entire leg part needs to be replaced. Never perform the replacement at site, for the assembly needed after replacement shall be asked to Ishida qualified service staffs.

1. Loosen each lock nut for the jack at the bottom part of legs.
2. Turn the jacks at four locations, to adjust the height.
3. Perform the conveyor horizontal adjustment by adjusting each jack, and tighten each lock nut for the jack.



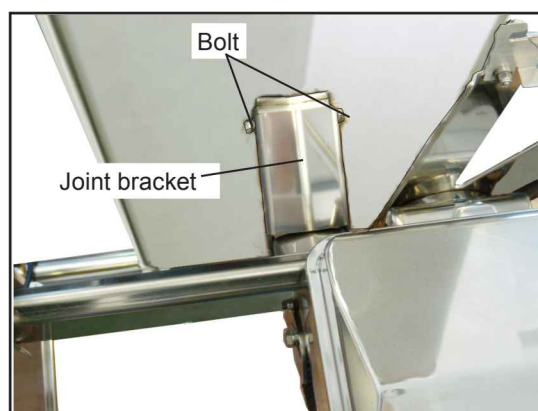
NOTE

It is adjustable in the range of 100 mm.

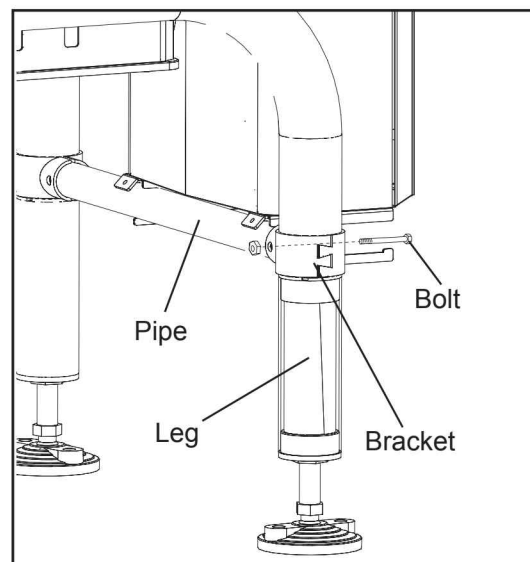
2.6.2 Extended leg model (optional)

Follow the steps below to change the conveyor height.

1. Remove the bolt of the connection fitting that fixes the electrical unit and the machine, at the electrical unit side.

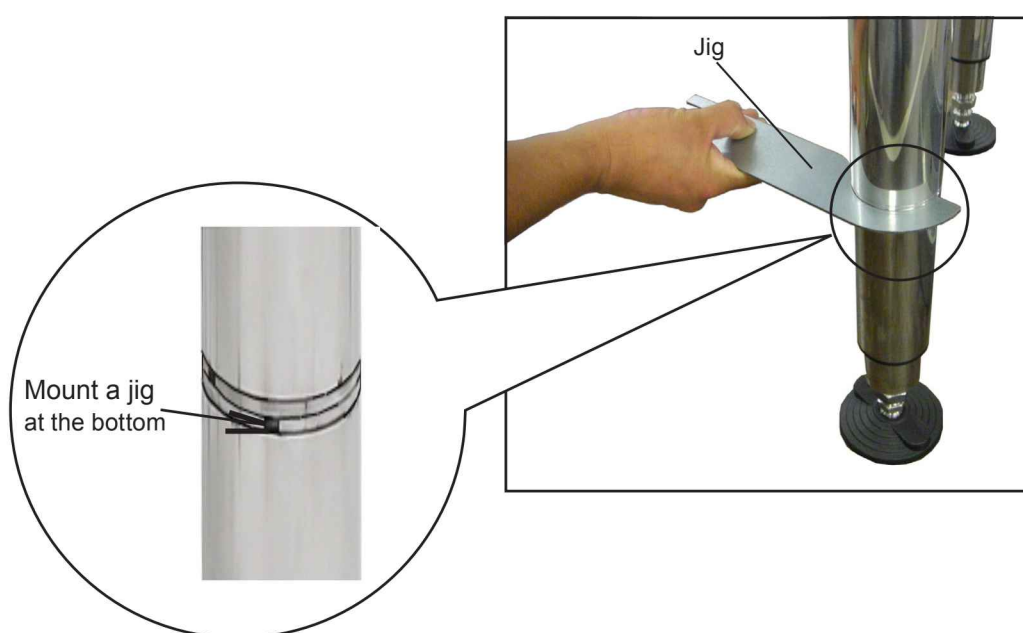
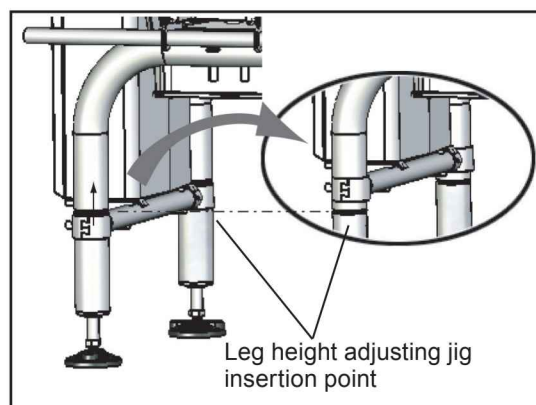


2. Loosen the right and left bolts fixing the pipe on which the electrical control unit is placed. Loosen the lock nuts fixing the bolts first.



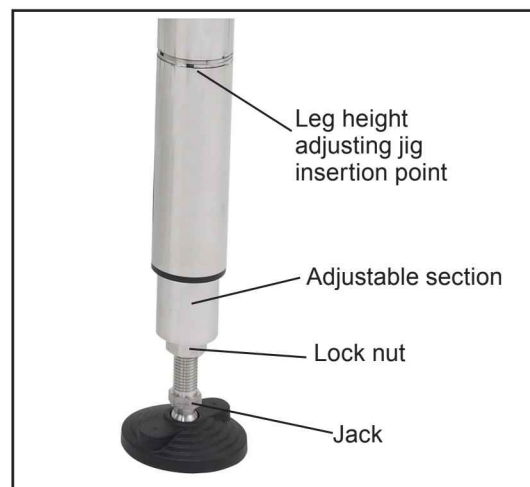
3. Fix a whole set of the electrical control unit, pipe, and brackets temporarily at the position higher than the leg height adjusting jig insertion point.

4. Attach the jig for adjusting height to the jig attachment part on the leg. Loosen the adjustable section by turning the jig in a clockwise direction as viewed from above.



2 SETUP

5. Loosen a lock nut attached to the jack located at the lower part of the leg on the adjustable section side.
6. Turn the adjustable section on the leg to adjust the height.
7. After changing height of all legs, fix the adjustable section of the legs by jig.
8. Replace the electrical control unit, which was moved temporarily, to the original position.
9. Fix the joint bracket removed at the procedure 1 by bolts to connect with the electrical control unit.
10. Adjust the levelling of the conveyor. Turn the jack to adjust with the loosened lock nut of the jack.
11. Tighten the lock nut of the jack.



NOTE

It is adjustable from the ground to the conveyor surface in the range of 700 mm to 1000 mm.

